SIMATS ENGINEERING

Assessment Tool 2 – Article Review

Course Name: Artificial Intelligence | Course Code: CSA17

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Weightage: 50% | Duration: 1 Hour | Questions: 4 | Total Marks: 40

Selected Article (peer-reviewed, published after 2019):

Abnoosian, K., Farnoosh, R., & Behzadi, M. H. (2023). Prediction of diabetes disease using an ensemble of machine learning multi-classifier models. *BMC Bioinformatics*, 24(1), Article 337.
DOI: **10.1186/s12859-023-05465-z** | Publisher: Springer Nature (BMC) | Open Access (CC BY 4.0)
Full text (PubMed Central): https://pmc.ncbi.nlm.nih.gov/articles/PMC10496262/

TASK 1 — Article Summary

(a) Full citation, domain and ML problem addressed

Citation: Abnoosian, K., Farnoosh, R., & Behzadi, M. H. (2023). Prediction of diabetes disease using an ensemble of machine learning multi-classifier models. *BMC Bioinformatics*, 24(1), 337. https://doi.org/10.1186/s12859-023-05465-z

Domain / application area: healthcare analytics — specifically computer-aided clinical diagnosis of diabetes mellitus using patient physical-examination and blood-test records.

Specific ML problem addressed: a **multi-class supervised classification** problem — assigning each patient record to one of three classes (Diabetic, Non-Diabetic, Predicted-Diabetic/pre-diabetic) using an **imbalanced** dataset, and doing so through a pipeline that combines several individual classifiers (k-NN, SVM, Decision Tree, Random Forest, AdaBoost, Naive Bayes) into a single AUC-weighted **ensemble model**.

(b) Summary of key objective and research gap

The authors set out to build a more reliable diabetes-prediction pipeline that copes with three problems that repeatedly limit medical ML models: scarce labelled clinical data, frequent missing attribute values, and severe class imbalance between diabetic and non-diabetic patients. Their central claim is that no single classifier handles all three problems well at once — some models (e.g. Naive Bayes) are fast but assume a data distribution that clinical variables rarely satisfy exactly, while tree-based models can overfit small, imbalanced samples. The research gap identified is the lack of a single end-to-end framework that jointly optimises pre-processing (missing-value imputation, normalisation, feature selection/dimensionality reduction) and model combination specifically for imbalanced, multi-class clinical data, rather than tuning a single classifier in isolation. To fill this gap, the authors propose a pipeline-based framework that systematically searches over pre-processing combinations for each individual classifier, tunes hyper-parameters via grid search and Bayesian optimisation, and then merges the best-performing classifiers into a weighted ensemble whose weights are the models' own Area-Under-ROC-Curve (AUC) scores — a metric chosen specifically because it remains meaningful under class imbalance, unlike raw accuracy. This directly targets the gap between single-model diabetes-prediction studies and a genuinely imbalance-aware, multi-class clinical pipeline.

TASK 2 — Methodology Analysis

(a) ML technique(s) used and dataset description

Techniques used: six individual classifiers were trained and compared — k-Nearest Neighbours (k-NN), multi-class Support Vector Machine (SVM), a Decision Tree classifier (DT, CART-style with Gini impurity), Random Forest (RF), multi-class AdaBoost, and Gaussian Naive Bayes (GNB). A One-vs-One (OVO) scheme extended each binary-style classifier to the 3-class problem. The authors then built a **weighted ensemble (EMLM)** that combines the confidence scores of the best individual models, using each model's OVO-AUC as its voting weight, and tuned every model's hyper-parameters with grid search and Bayesian optimisation inside a 5-fold cross-validation loop.

Dataset: the Iraqi Patient Dataset of Diabetes (IPDD), collected from 1,000 patients (565 male, 435 female, aged 20–79) during in-hospital physical examinations at a specialised endocrinology and diabetes hospital in Iraq. After removing 7 duplicate records, 993 samples remained, split into three highly imbalanced classes: Diabetic (837), Non-Diabetic (103), and Predicted-Diabetic (53). Each record has 11 attributes — gender, age, fasting blood sugar, blood urea nitrogen, chromium, cholesterol, triglycerides, LDL, HDL, BMI, and HbA1c. Preparation involved converting categorical attributes (gender, class label) to numeric codes, imputing missing values with k-NN imputation cross-checked by a specialist physician, min-max normalisation and z-score standardisation of continuous attributes, and feature selection/dimensionality reduction via MRMR, PCA, and ICA, compared across four separate experiments.

(b) Mapping to CO4 topics; one strength and one limitation

CO4 Topic	How the article maps to it
Inductive Learning	All six base classifiers are trained inductively from 993 labelled examples to generalise to unseen patients; the paper explicitly compares how different inductive biases (distance-based k-NN, margin-based SVM, impurity-based trees, probabilistic GNB) perform on the same data.
Decision Trees	A Gini-impurity CART-style Decision Tree is one of the six base learners; its depth, split criterion, and leaf-size hyper-parameters are explicitly tuned (Table 2 of the article), directly mirroring the course's decision-tree construction and pruning-style regularisation concepts.
Statistical Learning (Naive Bayes / Logistic-Regression family)	Gaussian Naive Bayes is used as the statistical baseline; the article shows its accuracy is highly sensitive to whether continuous features are standardised first — a direct illustration of how the Gaussian conditional-independence assumption interacts with pre-processing.
Reinforcement Learning	Not covered — the article is a one-shot (static) classification study with no sequential decision-making, states, actions, or reward signal, so it does not touch this CO4 pillar.

Methodological strength: the systematic, four-experiment comparison of pre-processing and feature-selection combinations (raw features vs. MRMR vs. PCA vs. ICA) for every classifier, combined with an AUC-weighted (rather than accuracy-weighted) ensemble, is a genuinely imbalance-aware design choice; using AUC as both the evaluation criterion and the ensemble-combination weight is methodologically coherent and well justified.

Methodological limitation: all experiments are evaluated purely by cross-validation on one single-hospital dataset — no external, held-out dataset from a different hospital or population is used to confirm the model generalises; combined with the very small minority classes (103 and 53 samples), this raises the risk that the reported near-perfect scores partly reflect fitting to this specific cohort rather than a generalisable diagnostic signal.

TASK 3 — Results & Critical Evaluation

(a) Key reported results

Best single-model average accuracy (AAC) after optimal pre-processing, from the article's Table 4:

Model	Best Avg. Accuracy (AAC)	Optimal pre-processing
k-NN	0.971 ± 0.003	Imputation + Normalisation, MRMR-6
SVM	0.948 ± 0.003	Imputation + Normalisation
Decision Tree	0.968 ± 0.003	Imputation + Standardisation, MRMR-10
Random Forest	0.988 ± 0.003	Imputation + Normalisation
Gaussian Naive Bayes	0.926 ± 0.006	Imputation + Standardisation + PCA-12
AdaBoost	0.961 ± 0.003	Imputation + MRMR-10

Best ensemble model (k-NN + AdaBoost + DT + RF, from the article's Table 9) vs. single models:

Metric	Best Ensemble (EMLM)	Best Single Model (RF)
Precision	0.986	— (RF single-model AAC only reported)
Recall	0.979	—
F1-Score	0.985	—
Accuracy	0.998	0.988
AUC-ROC	0.999	—

The authors also benchmark against a prior published hybrid model (k-NN + ID3 decision tree, by Soukaena Hassan et al.) that had reported 98.25% accuracy on a related Iraqi diabetes dataset; their proposed ensemble reaches 99.87% accuracy on the same style of comparison (article Table 10), which they present as their headline improvement over prior work.

(b) Critical evaluation

Are the reported metrics realistic? The reported figures (up to 99.8% accuracy and 0.999 AUC) are extremely high for a real clinical dataset, and should be read with caution rather than taken at face value. Several factors could inflate them: (i) the dataset is small (993 records) and highly imbalanced (84% diabetic), so even a naive majority-class predictor already scores well above 80% accuracy, which somewhat cushions the apparent accuracy gains, though the authors partly mitigate this by also reporting AUC; (ii) the pre-processing and feature-selection combination was chosen separately *per classifier* to maximise its own cross-validated score across four experiments — this is effectively many comparisons being made on the same data before the best combination is reported, which risks a mild form of selection/optimism bias in the headline numbers; (iii) cross-validation folds were drawn from the same single-hospital cohort, so the reported performance mainly reflects how well the model fits *this* population rather than how it would generalise elsewhere.

Potential biases: a clear dataset bias exists — all 1,000 patients came from one specialised hospital in Iraq, so the model's applicability to other ethnicities, healthcare systems, or diagnostic conventions is untested. There is also an evaluation-protocol bias: because different pre-processing pipelines were tuned per model using the same cross-validation data that ultimately produced the reported scores, there is no fully independent test partition, increasing the chance that the numbers are optimistic. The extreme minority-class sizes (53 pre-diabetic, 103 non-diabetic records) also mean per-class performance for those groups is statistically less reliable than the aggregate metrics suggest.

Additional experiments that would strengthen the findings: external validation on an independent, geographically distinct diabetes dataset (e.g. PIMA Indian, or a different-hospital Iraqi cohort); reporting per-class precision/recall (especially for the 53-sample pre-diabetic class) rather than only micro-averaged metrics, since aggregate scores can mask poor minority-class performance; a proper held-out test set that was never used during any hyper-parameter or pre-processing search; statistical significance testing (e.g. paired t-tests or confidence intervals across folds) when comparing the ensemble to Random Forest alone, since RF alone already reaches 0.988 AAC and the marginal ensemble gain may not be statistically significant given the added complexity; and a computational-cost/latency comparison, since the ensemble requires training and running six tuned models instead of one.

(c) Real-world applicability

The framework is best suited as a **clinical decision-support / triage tool** inside endocrinology or primary-care settings — flagging high-risk patients from routine blood-panel results (FBS, HbA1c, lipid profile, BMI) for earlier confirmatory testing, rather than replacing a physician's diagnosis. It could also support population-level screening programmes where lab data is already digitised. Challenges in scaling this to industry/hospital-wide deployment include: **data availability** — most hospitals do not have a curated, de-duplicated, imputed dataset like IPDD, and merging data across hospitals raises interoperability and consent issues; **compute and maintenance cost** — the full pipeline (six tuned classifiers, Bayesian hyper-parameter search, MRMR/PCA/ICA experiments) is far heavier to retrain and monitor over time than a single model, which matters for smaller clinics; and **ethical/regulatory issues** — deploying a model trained on one hospital's population to diagnose patients elsewhere risks systematic misdiagnosis for under-represented groups, and any diagnostic-support tool in healthcare typically requires regulatory clearance and clear human-in-the-loop safeguards before clinical use.

TASK 4 — Reflection & Learning Outcome

(a) Three key insights connected to CO4 concepts

- **Insight 1 – Inductive learning & inductive bias:** seeing six structurally different classifiers (distance-based, margin-based, tree-based, probabilistic) trained on the identical dataset and produce a spread of accuracies from 0.926 to 0.988 made concrete how much a model's *inductive bias* — its built-in assumptions about what a "simple" or "likely" hypothesis looks like — shapes generalisation, reinforcing the course's discussion of hypothesis-space bias in inductive learning.
- **Insight 2 – Decision Tree hyper-parameters as a form of pruning:** the article's tuned Decision-Tree configuration (Gini criterion, max_depth = 8, min_samples_leaf = 2, min_samples_split = 0.2) is effectively a pre-pruning strategy; connecting this to the course's pre-pruning vs. post-pruning discussion clarified that constraining tree growth *during* training via hyper-parameters achieves a similar overfitting-control goal to explicit cost-complexity post-pruning.
- **Insight 3 – Statistical learning assumptions matter more than the algorithm's label:** Gaussian Naive Bayes was the weakest single model overall, but improved specifically under z-score standardisation — a direct, dataset-level demonstration of the course concept that a statistical learner's performance depends heavily on how well the data satisfies its underlying distributional assumption (Gaussian conditional independence here), not just on the algorithm itself.

It is also worth noting, as a fourth reflection, that the article does not touch the Reinforcement Learning pillar of CO4 at all — underscoring that RL addresses a fundamentally different problem class (sequential decision-making under delayed reward) than the one-shot classification covered here.

(b) Proposed extension: novel experiment / enhancement

Proposed experiment: extend the IPDD-style dataset from a single cross-sectional snapshot into a **longitudinal, sequential-decision problem** by collecting repeat visits (e.g. every 3–6 months) for the same patients, and modelling disease-stage progression/treatment response as a **Markov Decision Process**: states = {Non-Diabetic, Predicted-Diabetic, Diabetic} refined by the existing 11 lab features; actions = {Monitor, Diet/Lifestyle counselling, Medication}; reward = clinically meaningful improvement (e.g. HbA1c/FBS moving toward the healthy range) minus a penalty for progression toward the Diabetic class. A Q-Learning or policy-gradient agent could then be trained to recommend the next check-up interval and intervention for a Predicted-Diabetic (pre-diabetic) patient, directly extending the article's static 3-class classifier into the CO4 Reinforcement-Learning pillar that it currently leaves untouched.

Feasibility: high — the same specialised hospital that supplied IPDD already performs periodic metabolic panels on patients, so the required longitudinal lab measurements are a natural extension of existing clinical workflow rather than a new data-collection burden; the three-class label structure the authors already use (Diabetic / Non-Diabetic / Predicted-Diabetic) maps cleanly onto MDP states without needing new annotation categories.

Expected impact: where the current model only answers "what class is this patient today", the proposed RL extension would answer the clinically more actionable question of "what should be done next, and when should the patient be re-checked" — shifting the contribution from one-off diagnosis support toward proactive, personalised diabetes-progression management, and giving the article's otherwise strong classification pipeline a natural bridge into the sequential decision-making side of CO4 that the original study does not address.

End of Article Review — AT2, Artificial Intelligence (CSA17), CO4. Source article: Abnoosian et al., BMC Bioinformatics 24, 337 (2023), doi.org/10.1186/s12859-023-05465-z (CC BY 4.0).